

School-Level Racial Achievement Correlations Across States*

Is Madison an Outlier? Replicating the Wisconsin School Scatter in Eight Other States

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Abstract

A striking finding from our Wisconsin analysis is that Black student proficiency in Madison Metropolitan School District (MMSD) schools is only weakly correlated with the White proficiency level of the same school (Pearson r of roughly 0.23), while the White–Hispanic school-level correlation is strong and universal. This companion report asks whether that pattern is unusual nationally. We collect school-level test results by race from the education departments of eight states — California, Texas, Illinois, New York, Ohio, Georgia, North Carolina, and New Jersey — and replicate the Wisconsin school-level scatter (White vs. Black and White vs. Hispanic proficiency, elementary/middle grades, pooled over the 2–3 most recent years) in each. To avoid over-reading noisy estimates, we report correlations only for places with at least 16 schools contributing both rates. Statewide White–Black school-level correlations cluster between 0.58 and 0.76 in every state, well above MMSD’s 0.23. The college-town districts we flag as MMSD analogs (Berkeley, Evanston, Ithaca, and others) are nearly all too small to clear the reporting threshold — itself evidence of how unusual MMSD’s combination of size and college-town character is. The one analog that does, Athens–Clarke County, GA (18 schools), shows a White–Black correlation of 0.33, well below its statewide 0.59, weakly consistent with a college-town pattern. White–Hispanic correlations are uniformly strong (0.6–0.8) in every reportable sample, including MMSD, mirroring the Wisconsin result.

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Introduction

The Wisconsin report documents that within MMSD, the school-level association between White and Black proficiency is remarkably weak: schools where White students post very high ELA proficiency are barely more likely to have high Black proficiency (Pearson $r \approx 0.23$ across 38 MMSD schools with both rates published). Statewide, that correlation is 0.46, and in Milwaukee it is 0.62. For Hispanic students the association is strong everywhere in Wisconsin, including MMSD ($r \approx 0.63$).

Two interpretations compete. One reads the weak MMSD correlation as a Madison-specific failure: whatever school-level resources and conditions lift White achievement in MMSD do not reach Black students. The other notes that MMSD is a high-SES college-town district — White proficiency is uniformly high and compressed across its schools, so there is little variation on the horizontal axis for Black proficiency to co-move with, and the Black student population may be more residentially and economically distinct from the White population than in a typical district.

Distinguishing these stories requires comparison districts. The Stanford Education Data Archive publishes school-level averages only for all students combined (race breakdowns exist only at the district level), so it cannot speak to within-district school-level racial correlations. We therefore went to state education department report-card downloads directly. Eight states publish school-level proficiency by race/ethnicity in downloadable form with usable recent coverage: California, Texas, Illinois, New York, Ohio, Georgia, North Carolina, and New Jersey. Together with Wisconsin, the analysis covers nine states, including four of the five largest U.S. school systems (New York City, Los Angeles, Chicago, Houston).

For each state we replicate the Wisconsin Figure 12 design: each point is a school; Panel A plots White (x-axis) against Black (y-axis) ELA proficiency, Panel B against Hispanic proficiency; points are colored to flag the state’s largest urban district (the Milwaukee analog) and a designated college-town district (the MMSD analog); the figure notes report Pearson correlations for the overall sample and each subgroup.

A critical caveat throughout: proficiency levels are not comparable across states. Each state administers its own test with its own cut scores — Texas “Meets Grade Level or Above” and NC “Grade Level Proficient” are not the same standard as Wisconsin’s Forward Exam proficiency. All cross-state comparisons in this report concern the within-state shape of the school scatter — the correlations — never the levels.

Data

Each state’s data come from its education department’s public report-card or assessment download site, harmonized to a common school-by-race panel (analysis/13_download_states.py, analysis/14_load_states.py). ELA is the primary subject, grades 3–8 (elementary/middle), pooled over the 2–3 most recent available years. Masked cells are treated as missing, never imputed. Where a state publishes tested counts, pooling is count-weighted; where it publishes only percentages (Illinois, Ohio), pooling is an unweighted mean over grade-year cells.

Table 1: Data sources by state. “Years” are pooled within state. See DATA.md for URLs and per-state notes.

State	Test (proficiency standard)	Years	School-level N by race?	Masking rule
WI	Forward Exam (Proficient+)	2015-16 to 2022-23	Yes	n < 10 suppressed
CA	CAASPP Smarter Balanced (Standard Met+)	2022-23 to 2024-25	Yes	n < 11 suppressed
TX	STAAR (Meets Grade Level+)	2022-23, 2023-24	Yes	small cells masked
IL	IAR (Levels 4–5)	2023-24, 2024-25	No	small cells starred
NY	NYS 3–8 Tests (Levels 3–4)	2022-23 to 2024-25	Yes	small cells suppressed
OH	Ohio State Tests (Proficient+)	2022-23 to 2024-25	No	“NC” for small cells
GA	Milestones EOG (Proficient + Distinguished)	2022-23 to 2024-25	Yes	too-few-students masked
NC	NC EOG (Grade Level Proficient)	2022-23 to 2024-25	Yes	bounded (<5, >95)
NJ	NJSLA (Met + Exceeded)	2022-23 to 2024-25	Yes	starred cells

Four additional states were attempted and set aside: Florida blocks scripted downloads

at the network level; Michigan’s MI School Data requires an interactive Power BI export; Minnesota’s data center timed out on repeated attempts; and Massachusetts’s DESE state reports require a session-based form export. These are documented in DATA.md for manual follow-up.

The designated largest-urban analogs (Milwaukee’s role in the Wisconsin figure) are Los Angeles, Houston, Chicago, New York City, Columbus, Atlanta, Charlotte-Mecklenburg, and Newark. The designated college-town analogs (MMSD’s role) are Berkeley, College Station, Evanston (CCSD 65), Ithaca, Athens (OH), Athens–Clarke County (GA), Chapel Hill-Carrboro, and Princeton.

Reporting threshold. A correlation computed over a handful of schools is dominated by noise (the standard error of r near zero is roughly $1/\sqrt{n-3}$, about 0.28 with 15 schools). Throughout this report we therefore report correlations only for places with at least 16 schools contributing both rates; smaller subsamples are shown as “—” with their school count. This screen removes nearly every college-town analog (and Newark, where heavy small-cell suppression leaves few White rates published): college towns are inherently small districts, and most have too few schools with published Black proficiency rates to support any correlation at all. That sparsity is itself informative about how unusual MMSD’s combination of size and college-town character is — Madison, at 38 schools with published Black rates, has no true peer in these eight states.

Results

Summary: where Wisconsin and MMSD sit

Table 2 and Table 3 collect the Pearson correlations from all nine state figures (cell entries are r with the number of schools in parentheses; “—” indicates fewer than 16 schools with both rates published, below the reporting threshold).

Table 2: School-level Pearson correlations, White vs. Black ELA proficiency.

State	Overall	College-town district	Largest urban district	Other
Wisconsin	0.46 (368)	0.23 (38) Madison	0.62 (38) Milwaukee	0.54 (292)
California	0.70 (938)	— (4) Berkeley	0.65 (63) Los Angeles	0.70 (871)

State	Overall	College-town district	Largest urban district	Other
Texas	0.59 (3,342)	— (15) College Station	0.56 (63) Houston	0.59 (3,264)
Illinois	0.72 (1,206)	— (15) Evanston	0.63 (132) Chicago	0.75 (1,059)
New York	0.62 (1,236)	— (5) Ithaca	0.44 (310) New York City	0.62 (921)
Ohio	0.76 (825)	— (1) Athens	0.62 (47) Columbus	0.76 (777)
Georgia	0.59 (1,778)	0.33 (18) Athens–Clarke	0.54 (57) Atlanta	0.62 (1,703)
North Carolina	0.58 (1,297)	— (11) Chapel Hill–Carrboro	0.45 (87) Charlotte–Mecklenburg	0.59 (1,199)
New Jersey	0.64 (551)	— (1) Princeton	— (3) Newark	0.64 (547)

Table 3: School-level Pearson correlations, White vs. Hispanic ELA proficiency.

State	Overall	College-town district	Largest urban district	Other
Wisconsin	0.61 (818)	0.63 (41) Madison	0.73 (40) Milwaukee	0.65 (737)
California	0.76 (3,914)	— (12) Berkeley	0.73 (200) Los Angeles	0.76 (3,702)
Texas	0.70 (4,914)	0.62 (16) College Station	0.68 (67) Houston	0.70 (4,831)
Illinois	0.77 (1,777)	— (14) Evanston	0.79 (156) Chicago	0.77 (1,607)
New York	0.71 (1,994)	— (9) Ithaca	0.62 (496) New York City	0.70 (1,489)
Ohio	0.75 (750)	— (0) Athens	0.60 (25) Columbus	0.74 (725)
Georgia	0.60 (1,767)	0.61 (18) Athens–Clarke	0.44 (57) Atlanta	0.62 (1,692)
North Carolina	0.53 (1,456)	— (13) Chapel Hill–Carrboro	0.50 (82) Charlotte–Mecklenburg	0.55 (1,361)

State	Overall	College-town district	Largest urban district	Other
New Jersey	0.77 (1,192)	— (5) Princeton	— (8) Newark	0.77 (1,179)

Four findings stand out.

1. The statewide White–Black school-level correlation is strong everywhere. Every state’s overall correlation falls between 0.58 (North Carolina) and 0.76 (Ohio) — Wisconsin’s 0.46 is actually the lowest of the nine states, pulled down in part by MMSD itself. Schools that produce high White proficiency generally also produce higher Black proficiency, across very different state contexts.
2. MMSD’s near-zero correlation is not replicated in any state’s overall sample, and almost no college-town analog is large enough to test. Every designated analog except Athens–Clarke County falls below the 16-school reporting threshold — MMSD, at 38 schools with published Black rates, is by far the largest district of this type in any of the nine states. Athens–Clarke (0.33, 18 schools) does show a markedly weaker White–Black correlation than its statewide sample (0.59), directionally consistent with a college-town pattern, but one district is thin evidence. The honest summary is that MMSD’s weak correlation is unmatched in any reportable sample, and the districts most similar to it are too small to say whether they share the pattern.
3. Largest-urban districts always show solid positive correlations. From New York City (0.44) to Los Angeles (0.65), the Milwaukee-style pattern — Black proficiency tracking White proficiency across schools — holds in every large urban district with sufficient data. The MMSD pattern is not an urban-district phenomenon.
4. The White–Hispanic correlation is strong and universal. In all nine states, and in every reportable subsample — including MMSD (0.63), Athens–Clarke (0.61), and College Station (0.62) — Hispanic proficiency co-moves with White proficiency at the school level, with correlations of roughly 0.6–0.8 throughout. Whatever decouples Black achievement from school-level conditions in MMSD does not decouple Hispanic achievement. This asymmetry, first documented for Wisconsin, appears to be a general feature.

Correlations at a glance

Figure 1 arrays the White–Black (red) and White–Hispanic (blue) correlations for every place — statewide samples and named districts — that clears the 16-school reporting threshold, ordered by the White–Black correlation. The only college-town analogs that qualify are Madison and Athens–Clarke County (College Station qualifies for the White–Hispanic panel only), and the figure’s message is stark: MMSD’s White–Black correlation of 0.23 sits at the bottom, with Athens–Clarke next at 0.33 and every other place at 0.44 (New York City) or above — while MMSD’s White–Hispanic correlation (0.63) is unremarkable and sits in the middle of the pack.

Interpretation for the MMSD boundary discussion

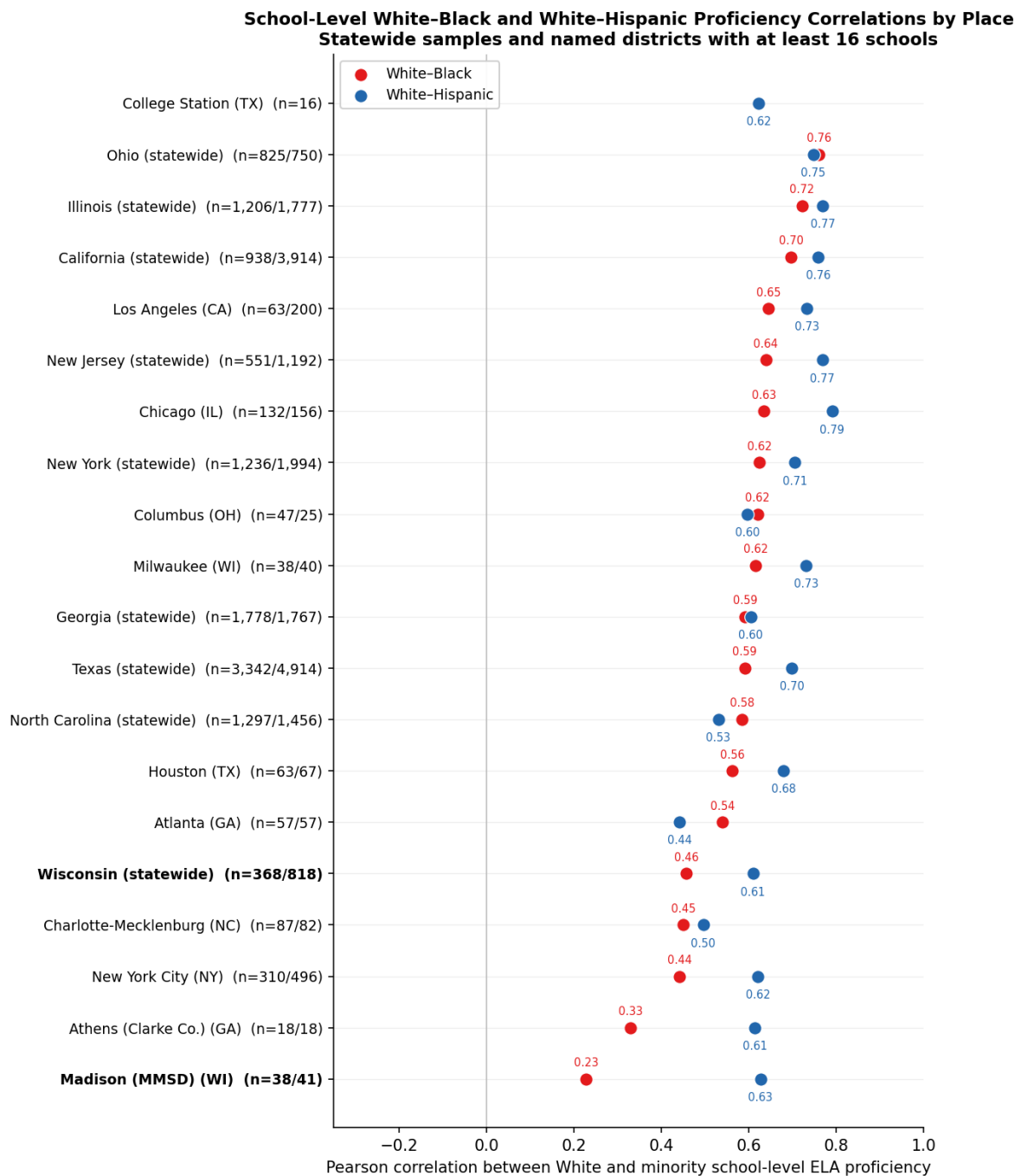
The cross-state evidence sharpens rather than dilutes the Madison finding. In every reportable sample — nine statewide samples and seven large urban districts — Black proficiency co-moves substantially with White proficiency across schools; MMSD’s 0.23 has no counterpart. Whether this reflects something specific to Madison or something generic to high-SES college-town districts cannot be settled here: the districts most similar to MMSD (Berkeley, Evanston, Ithaca, Chapel Hill-Carrboro) are simply too small to estimate the correlation reliably, and the one analog that clears the threshold, Athens–Clarke County, is only directionally suggestive. What can be said is that within MMSD, a school’s overall success (as proxied by its White proficiency) provides little information about how its Black students fare. For the boundary review, the implication parallels the main report’s within-school finding: reassigning students toward “higher-performing” schools — where performance is certified by overall or White proficiency — has a weak empirical basis for improving Black student outcomes in MMSD, because the school-level association those reassignment theories rely on is largely absent for Black students there.

State-by-state figures

Wisconsin appears first for reference; the remaining states follow in the order of Table 1. Axes are percentage proficient on each state’s own standard; the dashed line is parity.

Limitations

- Different tests and cut scores. Levels are not comparable across states; only within-state correlations and scatter shapes are compared. Even correlations can be affected



Note: Each row is a place; dots are Pearson correlations across schools within the place (ELA, grades 3–8, pooled recent years; see cross-state report for years and tests by state). Rows ordered by the White-Black correlation. Estimates based on fewer than 16 schools are omitted; the only college-town districts that qualify are Madison and Athens-Clarke County (omitted entirely: Athens (OH), Berkeley (CA), Chapel Hill-Carrboro (NC), Evanston (IL), Ithaca (NY), Newark (NJ), Princeton (NJ)). Correlations are within-state objects; proficiency levels are not comparable across states.

Figure 1: White-Black and White-Hispanic school-level correlations by place, ordered by the White-Black correlation. Only estimates based on at least 16 schools are shown.

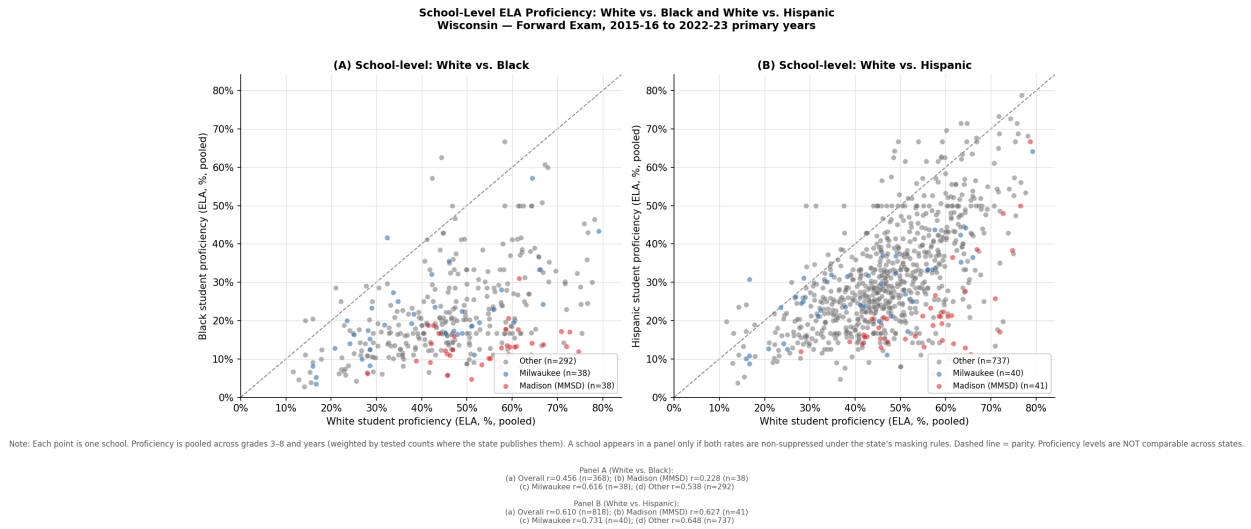


Figure 2: Wisconsin (Forward Exam, 2015-16 to 2022-23)

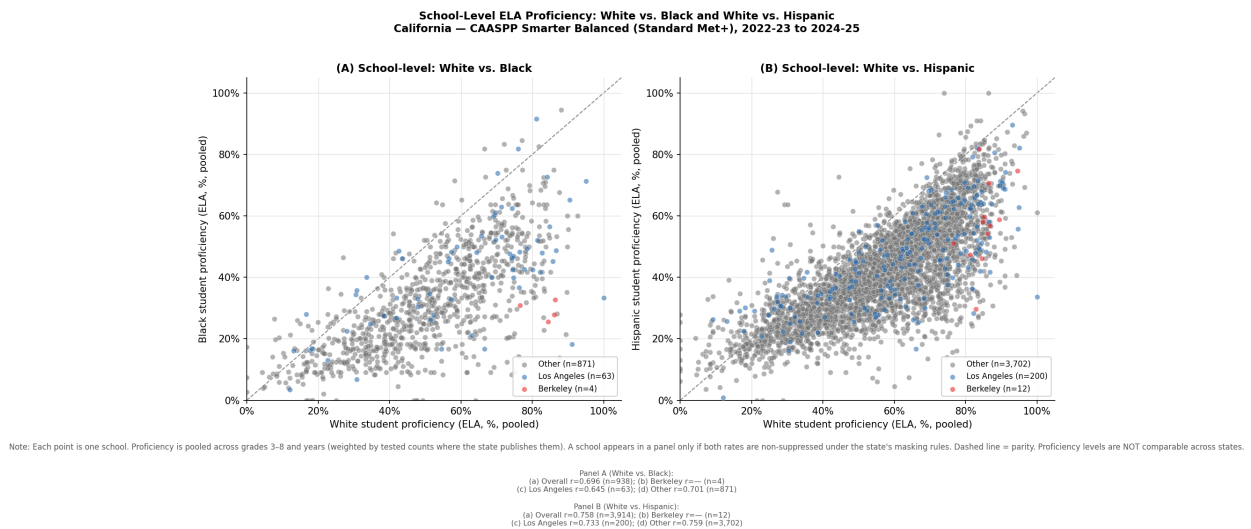


Figure 3: California (CAASPP Smarter Balanced, 2022-23 to 2024-25)

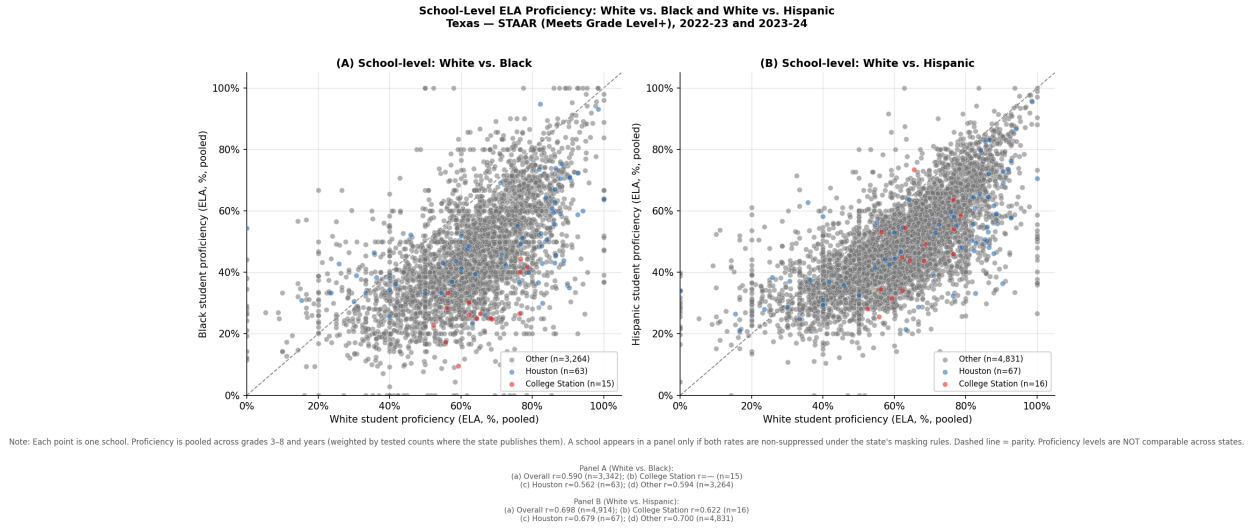


Figure 4: Texas (STAAR Meets Grade Level+, 2022-23 and 2023-24)

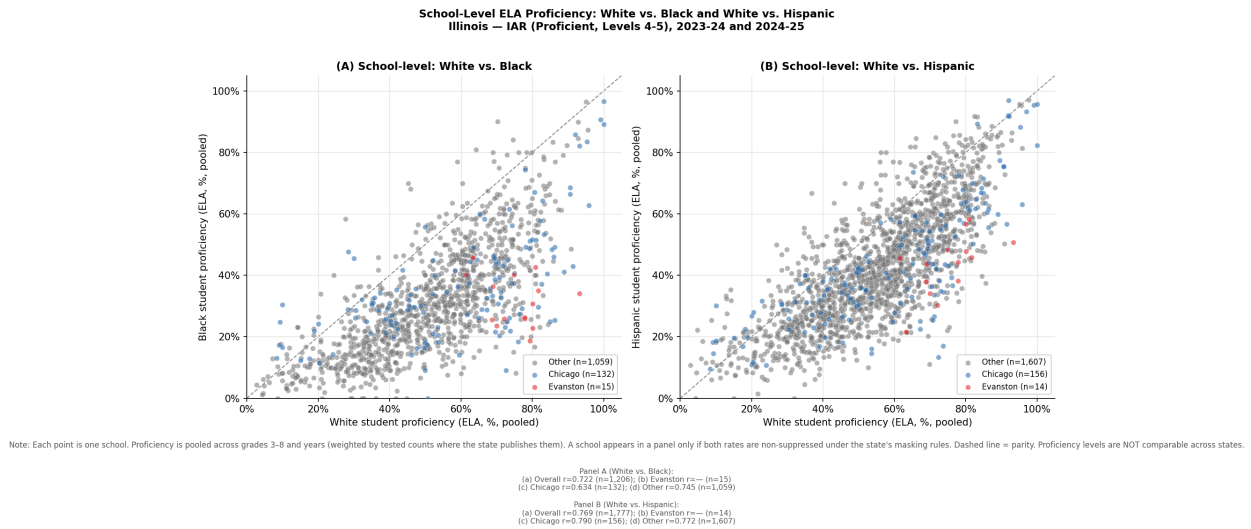


Figure 5: Illinois (IAR, 2023-24 and 2024-25)

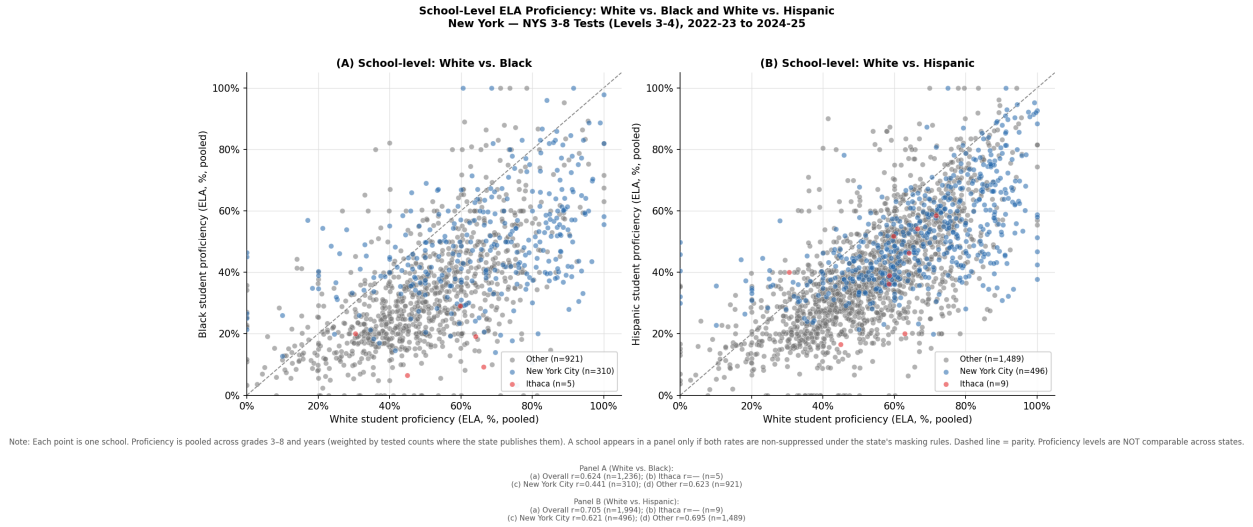


Figure 6: New York (NYS 3-8 Tests, 2022-23 to 2024-25)

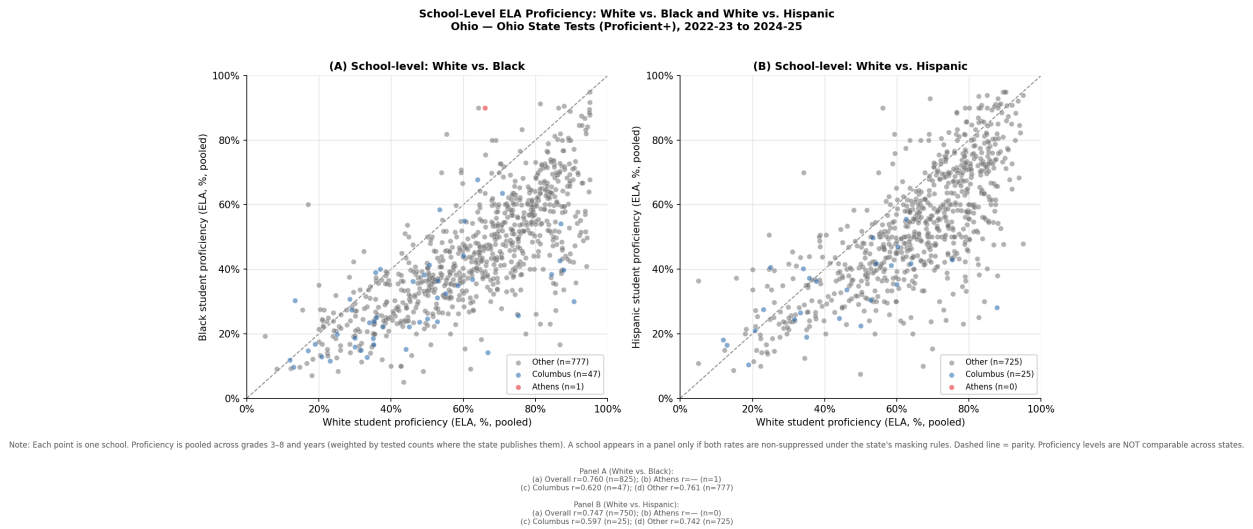


Figure 7: Ohio (Ohio State Tests, 2022-23 to 2024-25)

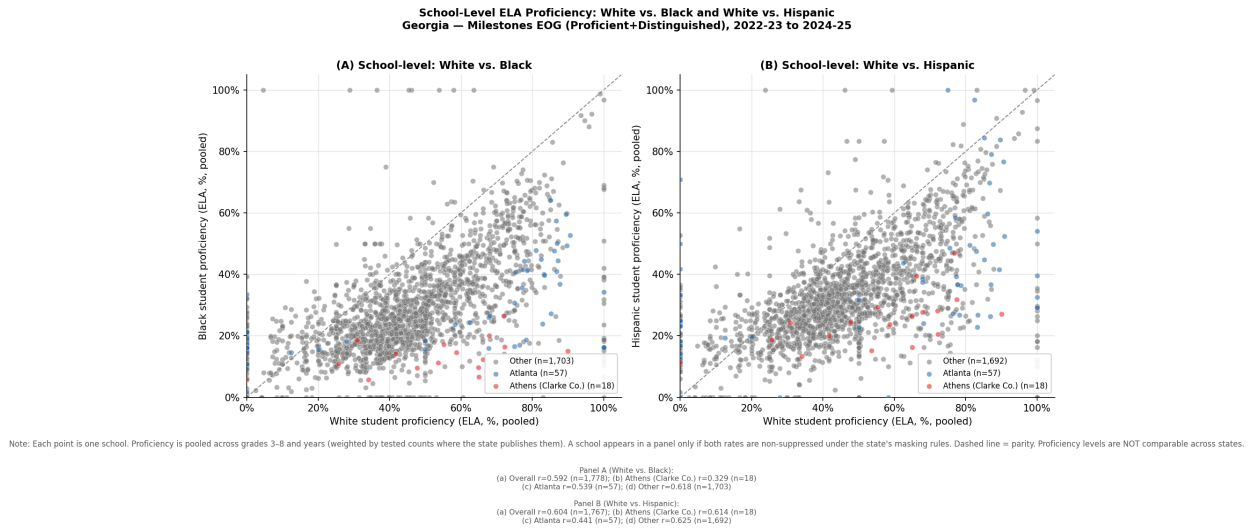


Figure 8: Georgia (Milestones EOG, 2022-23 to 2024-25)

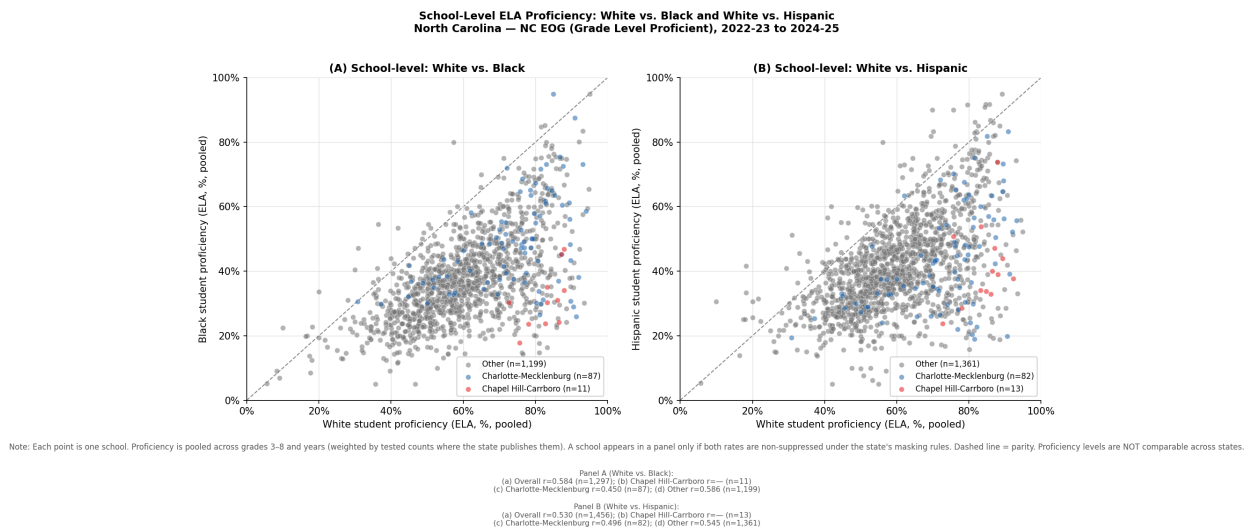


Figure 9: North Carolina (NC EOG, 2022-23 to 2024-25)

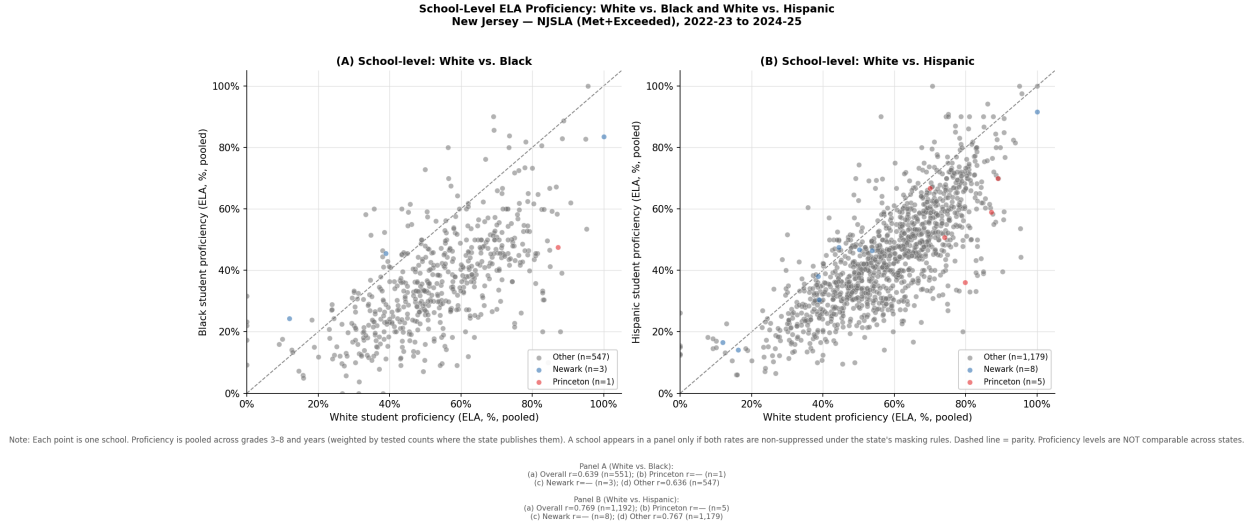


Figure 10: New Jersey (NJSLA, 2022-23 to 2024-25)

by where a state's cut score falls in the achievement distribution (a cut in the tail compresses variation and attenuates correlations).

- Masking regimes differ. Wisconsin suppresses cells under 10 students; California under 11; Ohio and Illinois suppress small cells without publishing counts; North Carolina bounds extreme percentages instead of suppressing. States with stricter masking systematically drop smaller-minority-population schools, which tends to raise observed correlations by retaining more diverse schools.
- Weighting differs where counts are unpublished. Illinois and Ohio pooled rates are unweighted means over grade-year cells rather than count-weighted; this adds noise for small schools.
- Charter treatment differs. Some states report charters inside their authorizing district (NYC geographic districts exclude most charters), others as their own districts (Texas, Georgia state charters). The “largest urban” subsamples are therefore not perfectly consistent in scope.
- College-town analogs are small. Only MMSD (38 schools) and Athens–Clarke (18) clear the 16-school reporting threshold in Panel A; every other designated analog has 15 or fewer schools with both rates published, so their correlations are not reported anywhere in this report. The college-town interpretation of the MMSD pattern therefore rests on one out-of-state district and remains a hypothesis.
- Years pooled differ by state (Table 1), spanning 2015–2023 for Wisconsin but 2022–2025 elsewhere. The Wisconsin correlations are essentially unchanged when restricted to recent years, so this is unlikely to drive the contrast.

Reproducibility

All code is in the repository: `analysis/13_download_states.py` (downloads), `analysis/14_load_states.py` (harmonization to `output/data/panel_school_race_multistate.parquet`, with a QC report at `output/tables/multistate_qc.txt`), and `analysis/15_multistate_school_scatter.py` (figures and `output/tables/multistate_school_correlations.csv`). State-by-state source notes, URLs, and quirks are documented in `DATA.md`.